Supply Chain Disruptions in Microelectronic Inputs: A Comparative Analysis of US and China

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1 Introduction

Microelectronic precursor materials are critical inputs in the global semiconductor value chain and play a central role in technological competitiveness, economic security, and geopolitical leverage [7]. Materials such as gallium, germanium, silicon, cobalt, and rare earth elements are indispensable for the production of semiconductors, power electronics, sensors, and defense-related technologies. Given their use in a wide range of strategic sectors (artificial intelligence, telecommunications, renewable energy, and advanced defense systems), supply disruptions at the precursor-material stage can propagate throughout the broader economy [8]. Yet existing methods to find bottlenecks and points of vulnerabilities are cognitively overwhelming given the number of materials.

Supply chains for many of these precursor materials are geographically concentrated and exhibit limited substitutability, creating structural dependencies that are vulnerable to export controls, trade restrictions, and geopolitical shocks [17]. These vulnerabilities are asymmetric across countries [14]. China's dominance in the processing and export of several refined precursor materials, combined with US dependence on a small number of foreign suppliers, shapes their relative bargaining power [17, 28].

This is evident from the Sankey diagrams of trade flows of precursor materials in Figure 1: China imports a lot of raw materials, but is a major exporter of many refined materials; whereas US imports very few raw materials but exports both raw and refined materials. Understanding lock-in and diversification at the material level is essential for evaluating the effectiveness of supply-chain resilience strategies [21]. Investments in alternative sourcing, domestic processing, and material substitution can significantly mitigate downstream risks but require long planning horizons and substantial coordination.

Across critical materials, supply chains range from partially independent to tightly interconnected, resulting in systems that compound vulnerabilities in non-obvious ways [6]. As a result, deriving system-level insight via analysis of individual networks is increasingly difficult as the number of materials to consider increases [7]. Existing analytical approaches often focus on individual components, bilateral dependencies, or certain segments of the value chain [32]. Network models and

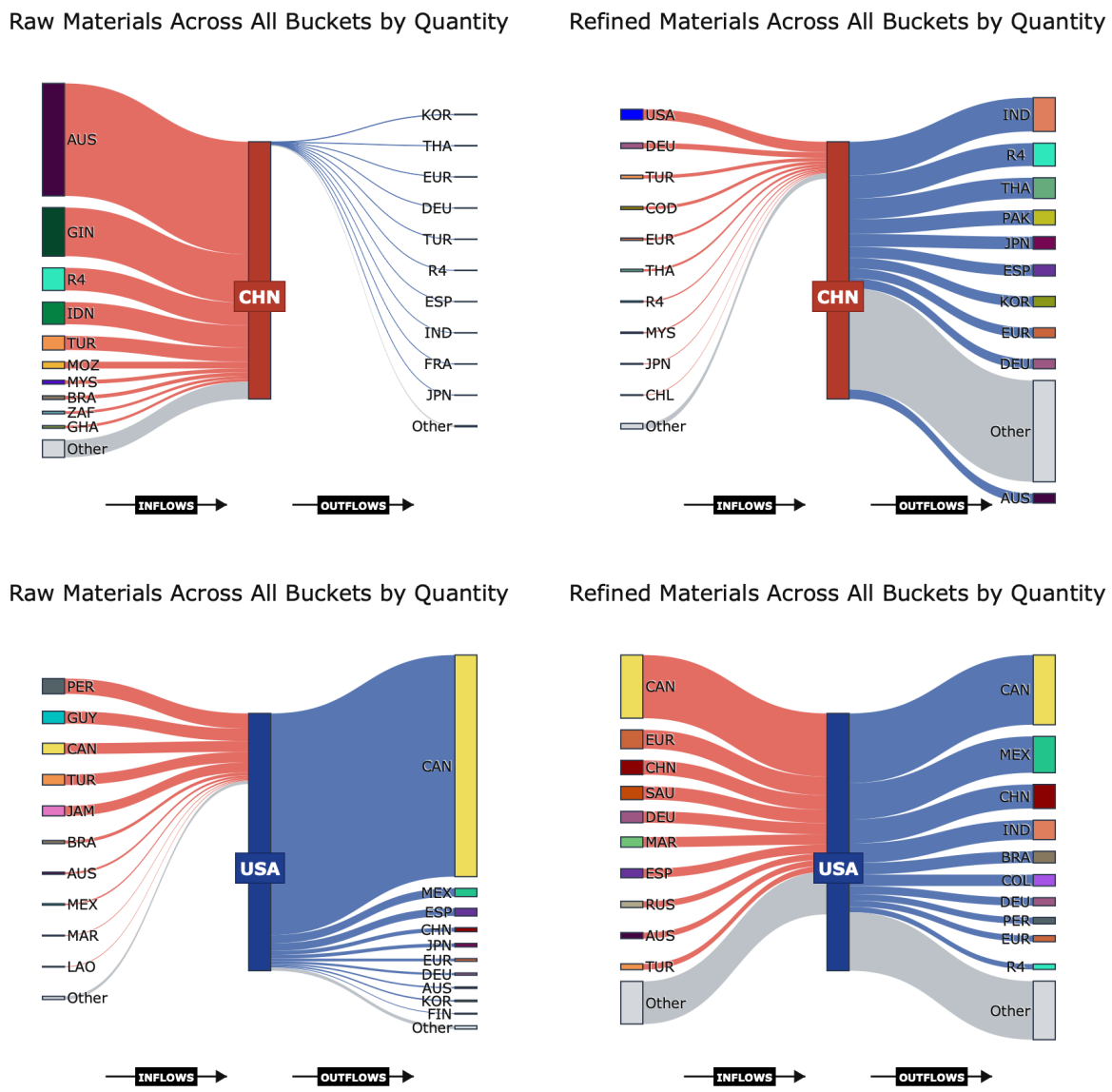

Figure 1: Total trade inflow of all materials (raw in the left column and refined in the right column) from top 10 countries and outflow to top 10 countries by China (top) and US (bottom) across years 2017-2025 by quantity. In each figure, the country (China or US) is shown in the center, and the importers and exporters are shown in the left and right, respectively. Thickness of links is proportional to the quantity of trade.

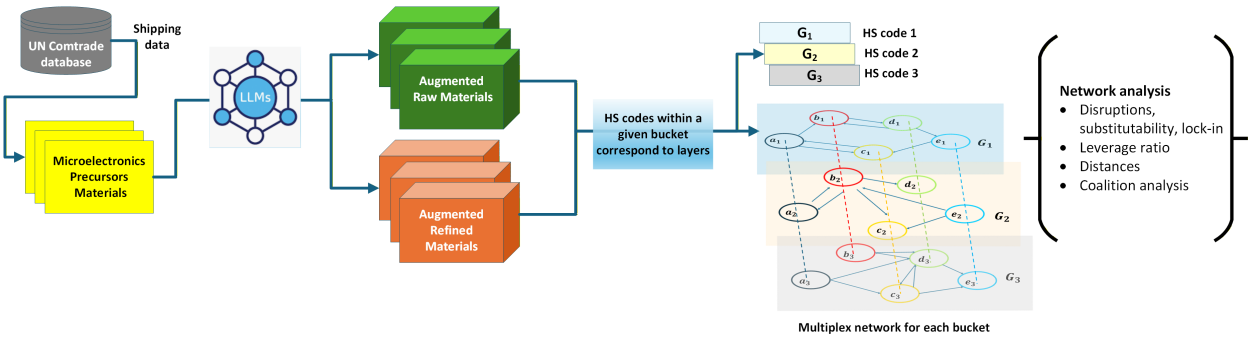

Figure 2: Overview of our approach: we determine an augmented set of precursors to microelectronics. For each bucket, we construct a multiplex network. An example multiplex network for a bucket is shown with layers G_1, G_2, G_3 (which correspond to the materials within the bucket). Each network G_i involves the same set of nodes $V = \{a, b, c, d, e\}$ (which correspond to countries), with the subscript denoting the layer. The edges might be different in each layer. **madhav:** Good figure please complete it. Say what the edges mean and their types.

analyses have been used to understand the complex dependencies that influence global trade and the key factors that drive them, e.g., [13, 20, 23, 26, 27, 30, 31]. These analyses have often been done on global trade networks constructed from datasets such as international trade (e.g., UN Comtrade and BACI, and GTAP) and economic information, such as those from WTO, National Bureau of Economic Research (NBER), World Integrated Trade Solution (WITS). As discussed in Liu et al. [27], different kinds of topological properties of networks have been used for analyzing trade networks, such as centrality metrics, distribution of distances, density and community structure. Centrality and clustering have often been used to formalize the fragility of trade networks [26, 31].

In this research, we study the comparative positioning of the United States and China with respect to their exposure to disruption in trade networks for precursors to advanced microelectronics; these are not captured in prior network analyses, which are mainly focused on understanding the characteristics of the global trade network as a whole. Further, prior work has considered networks for individual materials in isolation, which do not capture the numerous dependencies between these materials.

We introduce a structured framework for analyzing trade networks, enabling the identification of countries that act as critical bottlenecks across multiple commodities. We construct multiplex networks [9, 24] using the United Nations (UN) Comtrade Database [4], which provides detailed information on reporter and partner countries, Harmonized System (HS) codes (at the 6-digit level), trade values, trade flows, and transaction dates for global trade data. We consider precursor materials for advanced microelectronics, and partition them into broad target material classes (referred to as "buckets"), as shown in Table 1. Some of these materials were identified using generative AI tools [33] (described later in Section 4.2). We categorize these materials into raw and refined groups— these stages represent fundamentally distinct sources of supply chain risk and strategic leverage.

Figure 2 provides an overview of the approach and shows how dependencies are captured between materials. The network consists of a layer G_i for each distinct commodity in the bucket. Each node in G_i represents a country (so the set of nodes is the same for all the networks), and a

directed and weighted edge from one country to another signifies the exports, with the edge weight indicating the trade flow or value. A multiplex network provides a policy-relevant representation of strategic interdependence. Export controls, supply diversification, and industrial policy typically target specific materials or processing stages, not aggregate trade. By explicitly modeling these materials as layers, our framework allows analytical results to be aligned with real-world policy levers to help reduce systemic vulnerability.

We use three different metrics on these networks to measure the impact of trade disruptions. These metrics allow us to evaluate the diversification of a country's trade and detect strategic dependencies [11, 18]. By uncovering global trade patterns and vulnerabilities often hidden from conventional analyses, the approach allows direct comparison of suppliers that are critical to US and China. Numerous articles, e.g., [29], have noted that the US is unique in terms of strong partnerships with a number of countries, and that these collaborations could help improve resilience. We evaluate this hypothesis by considering metrics for coalitions of US and selected countries (Australia, Canada and Germany).

Disruptions in trade flows and lock-in. We model disruptions as the drop in trade flows and values for a given year when countries are hypothetically removed from the trading system; we identify the top k (1-3) supplier countries which lead to the maximum disruption to US and China. In the multiplex network setting, we consider the drop in flow at the level of a bucket, which is defined as the maximum drop in flow over all materials (HS codes) in the bucket. Table 1 shows the buckets and the associated HS codes. We base this modeling choice on the premise that interruptions in trade for any of these commodities can lead to significant disruptions for the entire basket of commodities. We construct a similar ranking of the top 1-3 disruptor countries for each bucket.

We use disruption sets to formalize lock-in in the following manner. We refer to the set of countries that appear in the set of the top 1, 2, or 3 most disruptive countries (over a time period) as the *supplier set* for a given country (US or China). *If the supplier set is small, we use it as an indication of lock-in.* We note this is different from other notions of substitutability that have been considered in the literature, e.g., [26], which are formalized in terms of human capital, instead of structure of trade relationships, as done here. Concretely, if the supplier market is very stable—such as in a system with high barriers to entry—and the two most dominant suppliers in any given year remain consistent across time, we will only see two countries appearing in the supplier set for $k = 2$. On the other hand, a trade ecosystem with more flexibility may see many different countries identified as the two most dominant suppliers for any given year. This would produce many more than two countries in the supplier set for $k = 2$.

Relative leverage in bilateral trade. The leverage ratio measures trade asymmetry and relative dependence in bilateral trade relationships [12]. This approach aligns conceptually with established frameworks in international trade and network analysis that quantify asymmetric vulnerabilities between trading partners [6, 22]. To evaluate the asymmetry and dependency structure in bilateral trade relationships, we calculate leverage ratios for individual commodities from both the US and China's perspectives. Leverage ratios are defined as the percent of imports for a commodity coming from a partner divided by the partner's percent of exports of that commodity going to the importer [12].

This ratio captures the relative dependence of one country on the other for a specific commodity. A

value greater than 1 indicates that the importing country is more dependent on the exporter than the exporter is on that trade flow, suggesting greater leverage or control for the exporting nation. Conversely, a value of less than 1 indicates the vulnerability of the exporter.

Impact of Coalitions. We analyze the leverage of country coalitions vis-a-vis China to assess whether strategic vulnerabilities can be mitigated through coordination. Coalition-based leverage ratios capture how bargaining power and dependence change when countries act collectively rather than individually, and they quantify the extent to which collective action can reduce structural vulnerabilities. Comparing leverage ratios across coalition configurations helps identify which partnerships can neutralize strategic advantages and which materials require deeper structural interventions beyond alliance formation.

Impact of Trade Disruption on Shortest Path Structures. The distribution of shortest paths in trade networks and the diameter have been used in prior literature for assessing their resilience, e.g., [26, 27]. Trade disruptions force rerouting of trade flows through more inefficient paths, thereby increasing transportation times, logistical costs, and the effective price of traded goods. As a result, paths between countries often lengthen or, in some cases, disappear entirely. Importers must find alternative suppliers through longer or costlier paths, which can increase congestion and vulnerability elsewhere in the system, as these substitute routes take on higher loads of trade volume. Over time, such pressure may increase the risk of cascading failures if additional links are lost. We measure changes in average distances and in number of nodes that become unreachable, to capture how the efficiency, resilience, and dependencies of international trade evolve under disruption. It highlights which countries or commodities are most vulnerable to supply interruptions and where diversification of trade routes or partners may be necessary to maintain stability.

2 Results

We use the United Nations (UN) Comtrade database for trade flows at the Harmonized System (HS) 6 level for the period 2017-2025. As mentioned earlier, we consider a target list of precursor materials for advanced microelectronics, which are partitioned into ten buckets. These materials are further categorized into raw and refined groups, as shown in Table 1. We also consider multiplex networks for each bucket. We discuss our results for the three metrics below.

Overall trade flow of raw and refined materials

To understand the big picture of trade flows, we show Sankey diagrams in Figure 1 which illustrate China's and the United States' trade flows in raw and refined materials across all buckets and all years combined. China's principal suppliers of raw materials include Australia, Guinea, R4, Indonesia, Turkey, and Mozambique [1]. Overall, China imports significantly more raw materials than it exports. In contrast, for refined materials, China's exports greatly exceed its imports. Among its relatively limited refined-material imports, the leading suppliers are the US, Germany, and Turkey, while India, R4, Thailand, and Pakistan constitute the major destinations for China's refined-material exports. Notably, the US does not appear among China's top ten importers of refined materials.

[1] R4 refers to the Association of Southeast Asian Nations (ASEAN) which include: Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand, and Vietnam.

The US exhibits the opposite pattern for raw materials: it exports substantially more than it imports. Its main export destinations for raw materials are Canada, Mexico, Spain, China, and Japan. For refined materials, the US imports the most from Canada, Europe, China, and Saudi Arabia, while its major export markets include Canada, Mexico, China, India, and Brazil.

It is important to note that the distinction between raw and refined materials is based on six-digit HS codes, and intermediate goods fall under the refined category. Note, in case of the US, Canada and China appear among both the top three importers and exporters of refined materials, indicating that the US holds a comparative advantage in certain refined products, while Canada and China dominate others.

2.1 Disruptions in trade flows and lock-in

We study the resiliency of multiplex networks through flow disruption. We take the total percentage reduction in the flow of a specific commodity after removing a pre-specified number of countries from the trade records. When the set of countries is one, we remove one country at a time from the records. For sets larger than one, we find the combination of countries that produces the largest reduction in flow when their impact is combined. Within each multilayered set of products, we identify the product and country list with the maximum disruption when 1, 2, or 3 countries are removed.

Figure 3 show the top disruptors for US and China for raw and refined materials. The heat map identifies the countries that act as the principal trade disruptors for each material bucket for both China and the US. For each year, a country is assigned a score of 1 if it ranks among the top three exporters (disruptors) for a given bucket, otherwise, a score of 0. The aggregate score is obtained by adding these yearly scores and can reach a maximum value of 27, corresponding to each k (k = 1, 2, 3) in the years 2017–2025. This aggregate score serves as a measure of the degree of dependency of China and the US, on their exporters.

For China, the leading disruptors in raw-material exports are Australia, Turkey, Guinea, Bolivia, and Canada. Australia attains an aggregate score of 25 for gallium, while Turkey records scores of 27 and 25 for boron and iridium, respectively. These high scores indicate that Australia and Turkey consistently rank among the top exporters in these buckets, suggesting a strong export concentration and a potential lock-in effect for China.

For the US, the principal exporters of raw materials are Peru, Turkey, and Canada, each achieving a score of 27 for boron, phosphorus, and silicon, respectively. Relative to China, the US appears to exhibit a higher overall degree of lock-in with respect to raw-material imports.

The contrast between the two countries is even more pronounced for refined materials. The US displays substantially greater lock-in than China in this category. For refined materials imported by the US, Germany (DEU) achieves a score of 27 for gallium, germanium, and silicon, while the UK records scores of 27 and 21 for cobalt and iridium, respectively.

EUR is a top exporter of Ga, Ge and Silicon to the US [2]. China is the dominant exporter of rare earth elements to the US, whereas US is the leading exporter of boron to China. Both countries also export cerium to one another, although they specialize in different HS code categories of cerium.

[2] Note that EUR represents European Union (EU) as a unit. Countries can trade individually as well as a part of EU.

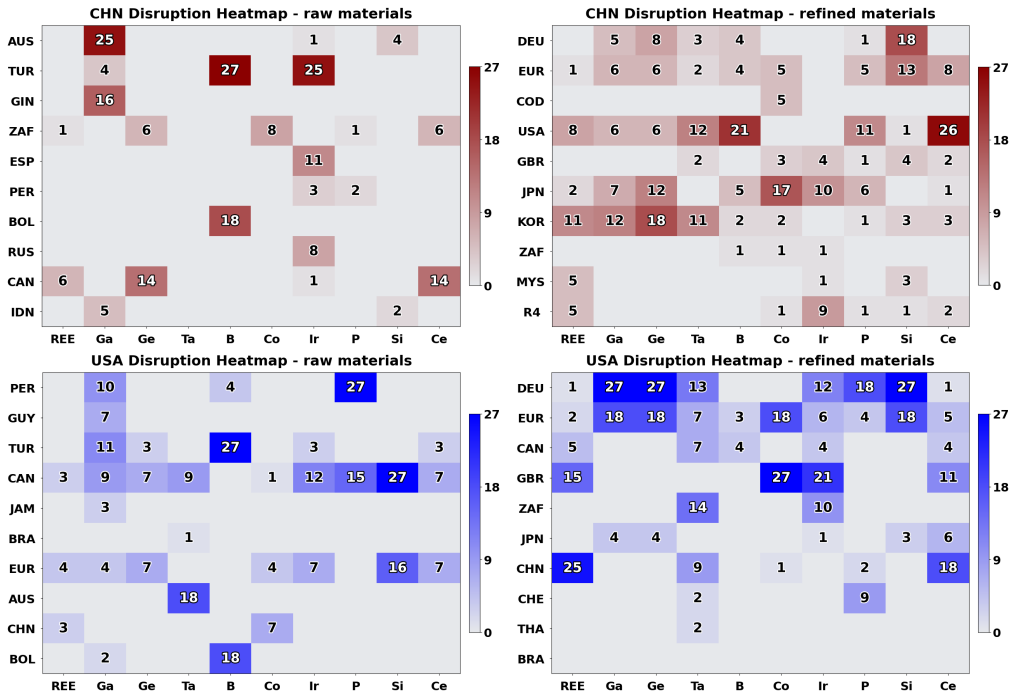

Figure 3: Disruption scores of countries for each bucket for US and China, for raw (left column) and refined (right column) materials, across years 2017-2025. The x-axis shows the bucket of commodities and the y-axis represents the countries who are the top disruptors for those buckets, for China (top row) and the US (bottom row). Each square states the disruption score.

The charts in Figure 3 collectively show that processing concentration, asymmetric dependencies, and limited redundancy define the vulnerability for microelectronic precursor materials. For the US, the greatest risk does not lie in access to raw resources but in the narrow chokepoints of global refining and high-value material processing, much of which is centered in China and other countries. For China, continued dependence on foreign sources for certain critical ores and technologies presents a strategic constraint.

Substitutability and lock-in

Our previous analysis quantified the percentage of trade flow lost when we remove top k exporters from the network. We conduct this analysis separately for each year and for each disruption size ($k = 1, 2, 3$) to identify the uniqueness of the top disruptors. If the same countries consistently emerge as top disruptors across years, and the number of unique countries identified is approximately equal to the disruption size (1, 2, or 3), this indicates persistent and highly influential actors over time and hence, low substitutability. However, if the number of influential countries is closer to the disruption size multiplied by the number of years in the sample, this suggests that the set of critical countries varies from year to year and hence there is high substitutability and low structural barriers to entry.

The tornado plot in Figure 4 shows the degree of substitutability and lock-in across different materials. It shows the number of distinct countries on which the importer depends. China is shown on the left and the US is on the right. The bars represent the number of distinct countries that

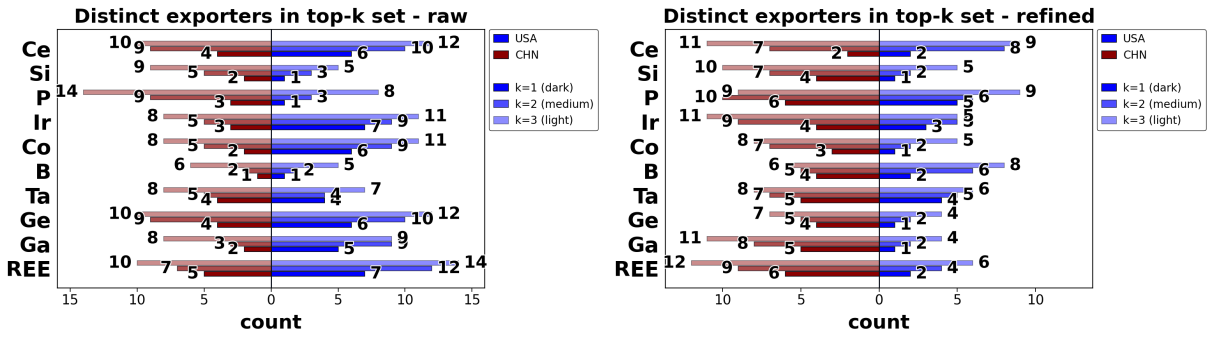

Figure 4: Number of unique countries that are in the top-k supplier set for the US and China, for raw (left) and refined (right) materials, across years 2017-2025, indicating the extent of lock-in.

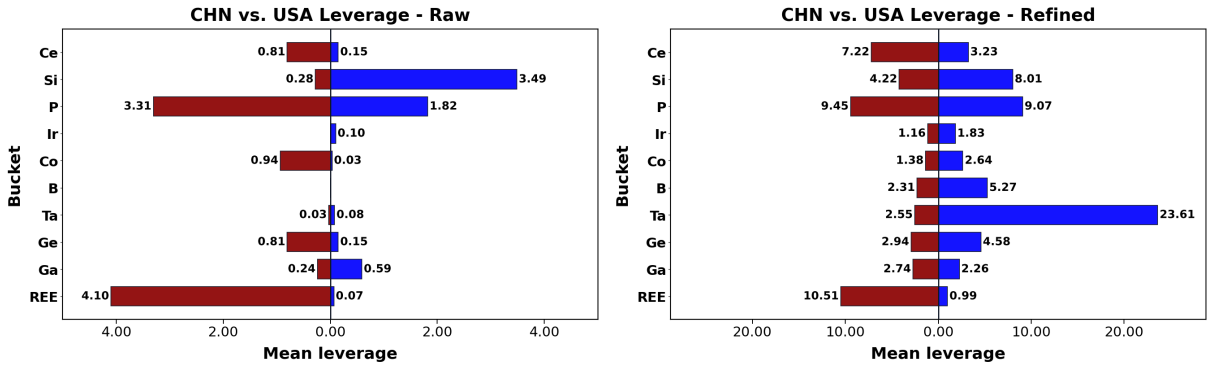

Figure 5: Leverage ratios (LR) for (left) raw and (right) refined materials. LR for China over the US are in red (to the left of the origin) and for the US over China are in blue (to the right of the origin) for different buckets (on the y-axis). LRs are calculated for each year and for each bucket, and then reported as the bucket-level mean across years (2017-2025).

appear among the top k exporters to US and China, and that could cause the greatest disruption to trade flows if removed from the network. A larger number of distinct countries indicates a lower degree of lock-in for the importing country. For each material, three bars are displayed, corresponding to k = 1, 2, and 3.

For both countries, the plots indicate that the US exhibits a higher degree of lock-in for refined materials than for raw materials. In contrast, China generally shows a higher degree of lock-in for raw materials than the US, with the exception of certain materials such as phosphorus and silicon.

The refined-materials plot further highlights that the US depends on a substantially smaller number of supplier countries than China, implying a higher degree of lock-in. For materials such as germanium, gallium, cobalt, and silicon, US supply is concentrated in only one or two countries. By comparison, China sources refined materials from a much broader set of countries, enabling greater scope for supplier substitution in the event of disruptions among top exporters. The narrower supplier portfolio of the US therefore implies more limited substitution options.

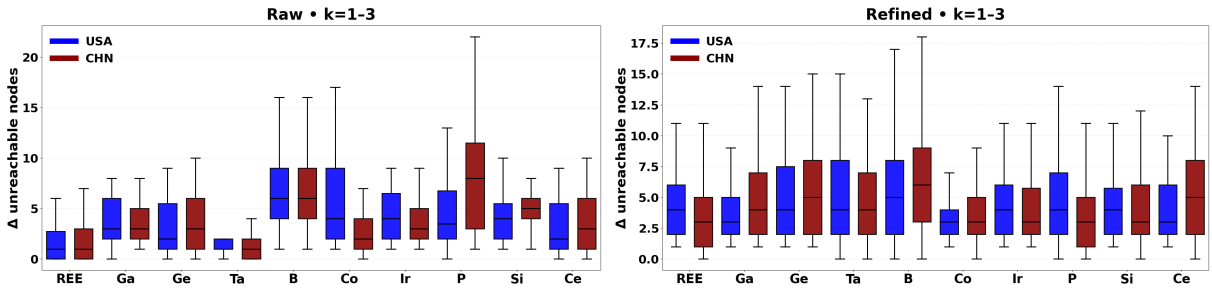

Figure 6: Box plots of increase in number of unreachable nodes (countries) for China and US, under restrictions of the maximum disruption set for $k = 1, 2, 3$ for each bucket of commodities of raw (left) and refined (right) materials.

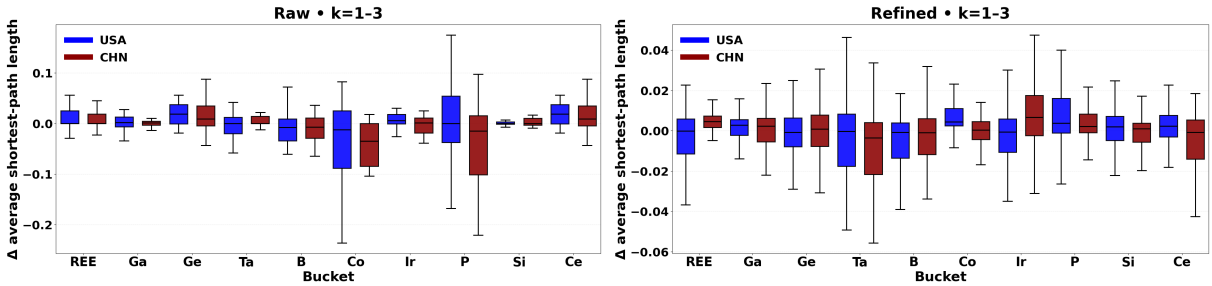

Figure 7: Box plots of changes in average shortest path distances from the suppliers which are reachable under restrictions of the maximum disruption set for $k = 1, 2, 3$, for China and the US, for each bucket of raw (left) and refined (right) commodities.

2.2 Leverage ratios

As mentioned earlier, leverage ratios are defined as the percent of imports for a commodity coming from a partner (by quantity) divided by the partner's percent of exports of that commodity going to the importer (by value); this is motivated by the study of leverage for critical pharmaceuticals [12]. This represents the asymmetry between the importer's dependence and low risk for the exporter. Figure 5 shows the relative leverage ratios of China and the US, for raw and refined materials. These are computed separately for the subset of raw and refined commodities in each bucket, and then averaged.

We observe that for some raw and refined materials China has high leverage over the US. In particular, for REE (both raw and refined), China has very high leverage over the US— it is over 4 for raw materials and over 10 for refined materials in the REE bucket. For a few other raw materials (Cerium, Phosphorous, Cobalt and Germanium), China has higher leverage ratios than the US. The leverage ratios for refined materials are, in general, much higher than those of raw materials. For many refined materials, the leverage ratios for certain materials in the buckets match for both US and China; for refined Silicon and Tantalum, the US has much higher leverage ratios than China. High leverage ratios in refined stages allow the exporting country to cause disruption or extract concessions with less risk of losing the export market.

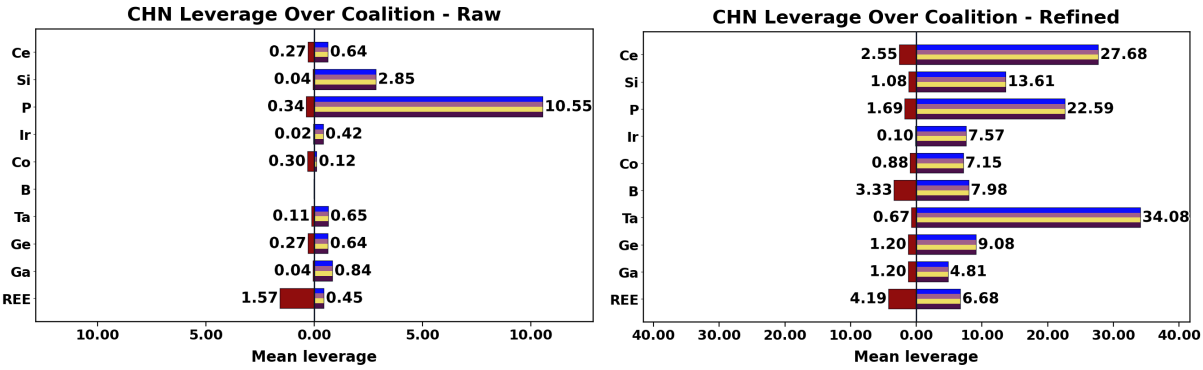

Figure 8: Leverage ratios of China over the coalition are in red (to the left of the origin) and the leverage ratio of the coalition (US, Canada, Australia plus Germany) over China, are in multicolors (to the right of the origin) for different buckets (on the y-axis) of raw (left) and refined (right) materials. LRs are calculated for each year and for each bucket, and then reported as the bucket-level mean across years (2017-2025).

2.3 Changes in shortest path structures

Trade restrictions on countries can change connectivity in the trade flow networks, and these changes need not be local. Further, some nodes might become unreachable if a subset is disrupted. Figure 6 present boxplots of the change in the number of unreachable nodes in the trade network following disruptions by the top k suppliers for ($k = 1, 2, 3$) for raw (left) and refined (right) materials, comparing the US and China. The metric captures the extent of network fragmentation induced by supplier removal, with larger values indicating greater vulnerability.

For raw materials, the distribution of impacts differs across materials, but China often exhibits comparable or higher disruption effects than the US, particularly for materials such as phosphorus, germanium and REEs. These results point to stronger lock-in at the raw-material stage for China, where supplier concentration remains high and substitution options are limited.

For refined materials, the US generally experiences larger increases in unreachable nodes than China across several materials, indicating higher exposure to disruptions among top suppliers. This pattern suggests a more concentrated supplier structure for the US in refined inputs. In contrast, China exhibits lower median impacts and narrower distributions for most refined materials, consistent with a more diversified supplier base.

We also compute the average shortest path distances from nodes which are connected to the US and China (Figure 7). Unlike the number of nodes from which a target country is not reachable, the average shortest path distances need not be monotone, and the differences are negative in some cases. This is because the averages are normalized by the number of nodes which are connected, which can vary, creating instances where the change in the average is negative.

The raw materials plot in Figure 7 exhibits substantially larger and more heterogeneous changes in average shortest-path length, particularly for cobalt and phosphorus. However, for refined materials, disruptions generally lead to modest changes in average shortest-path length for both countries, with effects centered close to zero. China's distributions are comparatively tighter, consistent with a more redundant and interconnected refined-material trade network.

2.4 Improved leverage ratios through coalitions

The results so far indicate that China has high leverage over the US with respect to many materials, and more resilient supply chains, especially for many refined materials. Numerous articles, e.g., [29], have noted that the US is unique in terms of strong partnerships with a number of countries, and that these collaborations could help improve resilience. We support this hypothesis by considering metrics for coalitions of US and selected countries. In Figure 8, we show the leverage ratio for a coalition consisting of US, Canada, Australia and Germany vs China for raw materials. In contrast with Figure 5, we find that the coalition has significantly higher leverage over China for all refined materials, and for most raw materials (other than Cobalt and Rare Earth Elements). Further, the leverage ratios of the coalition for refined materials are significantly higher than those for raw materials. Even for raw forms of Rare Earth Elements and Cobalt, China's leverage over the coalition is significantly reduced. In Section 6.2 of the Appendix, we show leverage ratios of smaller coalitions of the US with subsets of Canada, Australia and Germany. Even smaller coalitions show significant benefits. For instance, for Rare Earth Elements, China's leverage drops by about half for raw materials.

We also perform a temporal analysis of the leverage ratios which show that China has persistent leverage over the US in raw REE, Cobalt and Phosphorus whereas US leverage over the years in raw materials has been generally weak, except for silicon and phosphorus. Some refined materials show persistent leverage across multiple years, such as China's leverage over the U.S. in refined REE and phosphorus, and U.S. leverage over China in refined tantalum. They reflect structural vulnerabilities that cannot be addressed through short-term adjustments or market rebalancing. Even 4-country coalitions for the US show that persistent leverage of China in raw-materials cannot be resolved through alliances alone. Although temporal patterns show that coalitions can be very effective in case of refined materials. See a more detailed temporal analysis in the Appendix.

3 Discussion

Prior literature and our results show that trade restrictions have very complex effects. There are significant dependencies between materials used in industrial processes, which cannot be identified by examining them separately. Further, trade restrictions can have non-local impacts. The structured network framework we introduce here, including network metrics and multiplex networks provide a powerful approach for analyzing trade networks and the impact of such disruptions.

Our analysis highlights that the US and China have developed distinct supply chain networks for precursors of advanced microelectronics. One of the most striking differences is in the import and export of raw and refined materials. China imports significantly more raw materials than it exports. This is reverse for refined materials. The pattern is quite different for the US, which exports substantially more raw materials than it imports. The US also exports a lot of refined materials.

Refinement of many critical commodities, such as rare earth elements, cerium, gallium, germanium, and silicon intermediates, is more concentrated (than raw materials) in a small number of countries; often China is the most dominant or even sole supplier of some of these commodities. This contrasts with many raw materials, which are often sourced from geographically diverse producers such as Australia, Chile, or the Democratic Republic of Congo, but their refinement remains highly concentrated.

Another significant difference in the supply chain networks for the US and China is the degree of lock-in with respect to suppliers of different materials. China has a larger and diverse set of suppliers relative to the US, for refined materials in almost all the buckets we study, indicating greater substitutability and robustness in sourcing these materials. This contrasts with raw materials, for which China exhibits a slightly higher degree of lock-in than the US, though the difference is somewhat less than for refined materials. This suggests that for the US, the greatest risks are not access to raw resources, but the narrow chokepoints of global refining and high-value material processing, much of which is centered in China and other countries. For China, continued dependence on foreign sources for certain critical ores and technologies presents a strategic constraint.

Leverage ratios provide further insights into risks for importers. Both the US and China have high leverage with respect to some critical materials. For some buckets, such as Rare Earth Elements (REE), both raw and refined, China has a very high leverage over the US. On the other hand, US has high leverage over China for a few refined materials, such as refined Silicon and Tantalum. The reachability analysis shows both countries have similar vulnerabilities – restrictions on a small set of critical countries can disrupt connectivity with a similar number of countries. Our results highlight the complex effects of trade disruptions, and show that ensuring their resilience is crucial for global supply chains. We also find that the leverage of the US for both raw and refined materials can be improved significantly through close cooperation with other countries.

A more detailed coalition analysis (as shown in the appendix) reveals a clear contrast between raw and refined materials in terms of leverage and strategic vulnerability between the US and China. For raw materials, China retains substantial leverage over the US across several critical inputs, including cerium, phosphorus, cobalt, germanium, and REEs.

While coalition formation improves the US position, this leverage is only partially mitigated. Bilateral partnerships, such as those with Australia, Germany, or Canada, can eliminate or reduce China's advantage for specific raw materials, yet even a broad four-country coalition is insufficient to fully offset China's leverage in REE and cobalt. This persistence highlights the structural concentration and limited substitutability of extraction of upstream resources.

In contrast, leverage in refined materials proves far more sensitive to coalition formation. Relative to unilateral action, coalitions dramatically enhance the US position, and even limited partnerships can significantly erode China's leverage. Bilateral coalitions with Germany or Australia eliminate China's advantage in nearly all refined materials, with the exception of REE.

Importantly, a trilateral coalition involving the US, Australia, and Germany is sufficient to neutralize China's leverage entirely, including in refined REEs. Together, these findings underscore a fundamental asymmetry across the value chain: China's leverage is more established at the raw-material stage, whereas leverage in refined materials is more responsive to coordinated action.

Coalition leverage ratios provide direct policy relevance. They offer a systematic way to assess the effectiveness of international coordination, guide the selection of strategic partners, and prioritize investments in diversification, substitution, or domestic capacity where coalitions alone are insufficient.

Our analysis simplifies a number of factors and collaborations between countries, which might significantly alter the conclusions. We do not consider exogenous factors, such as the pandemic, wars and environmental disasters, which might have a bearing on both production and shipping

between countries. We also do not consider changes in demand patterns across the world, which might have a significant impact on trade.

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4 Data and Methodology

4.1 Dataset

We use the UN Comtrade database [4]. This dataset provides detailed trade flows by product (at an Harmonized System (HS) code level), including the trading countries, quantity and value. This data covers over 99% of world trades, as per their description [4]. The specific set of materials we study are listed in Table 1, along with their HS codes. These are further split into raw and refined materials. The methods used for selecting these materials, and for partitioning them as raw or refined are described in Section 4.2.

Table 1: Augmented set of critical materials organized as buckets, and associated HS6 codes used in the study. Each material is labeled as raw or refined.

Commodity	Type	HS6	Short Description
Cobalt (CO)			
	Raw	260500	Cobalt ores and concentrates
	Refined	282200	Cobalt oxides and hydroxides
	Refined	282739	Cobalt chlorides
	Refined	283329	Cobalt sulfates
	Refined	283699	Cobalt carbonates
	Refined	291529	Cobalt acetates and other acetates
	Refined	810520	Cobalt mattes, unwrought cobalt, powders
	Refined	810530	Cobalt waste and scrap
	Refined	810590	Other cobalt articles
Boron (B)			
	Raw	252800	Natural borates and concentrates
	Refined	281000	Boric oxide and boric acids
	Refined	281290	Other boron halides or halide oxides
	Refined	282690	Other inorganic fluorides (incl. boron trifluoride)
	Refined	283990	Other silicates (incl. borosilicates)
	Refined	284011	Disodium tetraborate (refined borax)
	Refined	284019	Other borates
	Refined	284020	Peroxoborates (perborates)
	Refined	284520	Boron isotopes and compounds
	Refined	284990	Other carbides (incl. boron carbide)
	Refined	285000	Hydrides, nitrides, azides, silicides, borides
Germanium (GE)			
	Raw	261790	Other ores and concentrates (germanium-bearing)
	Refined	281219	Germanium tetrachloride and other chlorides
	Refined	281290	Other germanium halides and halide oxides
	Refined	282560	Germanium oxides and hydroxides
	Refined	285000	Hydrides, nitrides, azides, silicides (germanides)
	Refined	811292	Germanium and articles; waste and scrap
Iridium (IR)			
	Raw	261690	Precious-metal ores and concentrates, other

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Commodity	Type	HS6	Short Description
	Refined	284390	Precious-metal compounds, other (incl. iridium compounds)
	Refined	381512	Supported catalysts with precious metal
	Refined	711019	Platinum-group metals, unwrought (incl. iridium), other
	Refined	711041	Platinum-group metals, in powder form
	Refined	711049	Platinum-group metals, other unwrought forms
	Refined	711292	Waste and scrap of platinum-group metals
Phosphorus (P)			
	Raw	251010	Natural calcium phosphates, unground
	Raw	251020	Natural calcium phosphates, ground
	Refined	280470	Phosphorus, white or yellow
	Refined	280910	Diphosphorus pentaoxide
	Refined	280920	Phosphoric acid
	Refined	281212	Phosphorus trichloride
	Refined	281213	Phosphorus oxychloride
	Refined	281214	Phosphorus pentachloride
	Refined	281390	Sulfides of nonmetals (incl. phosphorus sulfides)
	Refined	283510	Sodium phosphates
	Refined	283522	Potassium phosphates
	Refined	283524	Calcium hydrogenphosphates and phosphates
	Refined	283525	Ammonium phosphates
	Refined	283529	Other polyphosphates
	Refined	283531	Sodium triphosphate
	Refined	283539	Other polyphosphates, n.e.s.
	Refined	285390	Other inorganic compounds, n.e.s.
	Refined	291990	Other phosphoric esters and their salts
	Refined	310319	Mineral or chemical phosphatic fertilizers, other
	Refined	310530	Diammonium hydrogenorthophosphate (DAP) fertilizer
	Refined	740500	Copper master alloys (phosphor copper)
Silicon (SI)			
	Raw	250510	Silica sands and quartz sands
	Refined	280461	Silicon containing by weight $\geq$ 99.99% silicon
	Refined	280469	Other silicon
	Refined	281122	Silicon tetrachloride
	Refined	283911	Sodium metasilicates
	Refined	283990	Other silicates
	Refined	284920	Silicon carbides
	Refined	285000	Hydrides, nitrides, azides, silicides
	Refined	293190	Other organo-silicon compounds
	Refined	391000	Silicones in primary forms
	Refined	720221	Ferro-silicon, $\leq$55% silicon
Gallium (GA)			
	Raw	260600	Aluminum ores and concentrates (bauxite, gallium-bearing)
	Raw	260800	Zinc ores and concentrates (gallium-bearing)

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Commodity	Type	HS6	Short Description
	Refined	281219	Gallium trichloride and other chlorides
	Refined	282590	Gallium oxides and hydroxides
	Refined	283329	Gallium sulfates
	Refined	283429	Gallium nitrates and nitrites
	Refined	285000	Hydrides, nitrides, azides, silicides
	Refined	285390	Other inorganic compounds, n.e.s.
	Refined	381800	Doped semiconductor wafers and discs
	Refined	811292	Germanium; articles and scrap (semiconductor related)
	Refined	811299	Gallium and other specified metals; articles, n.e.s.
Rare Earths (REE)			
	Raw	261220	Monazite and other thorium/rare-earth ores
	Raw	261790	Other ores and concentrates (rare-earth bearing)
	Refined	280530	Rare-earth metals; scandium, yttrium
	Refined	284690	Compounds of rare-earth metals, n.e.s.
Cerium (CE)			
	Raw	261790	Other ores and concentrates (cerium-bearing)
	Refined	280530	Rare-earth metals (incl. cerium)
	Refined	284610	Cerium compounds
	Refined	360690	Ferro-cerium and other pyrophoric alloys
Tantalum (TA)			
	Raw	261590	Niobium, tantalum ores and concentrates
	Refined	282590	Tantalum oxides and hydroxides
	Refined	284990	Other carbides (incl. tantalum carbide)
	Refined	285000	Hydrides, nitrides, azides, silicides (tantalides)
	Refined	810320	Tantalum waste and scrap
	Refined	810330	Tantalum powders
	Refined	810391	Tantalum bars, rods, profiles
	Refined	810399	Tantalum, other forms

4.2 Methodology

Our methodology consists of two components. First, we select a set of critical materials related to advanced microelectronics. We augment this set, and partition them into raw and refined materials, using an LLM based approach. Next, we consider network models from trade flows and compute different network based metrics. We describe these components below.

4.2.1 Choosing a set of critical materials

We select a set of critical materials related to advanced microelectronics. We start with a set of core materials, such as Silicon, Gallium and Rare Earth Elements. We identify other materials which are representative of the primary industrial forms of the target materials using an LLM based approach [33]; the augmented set for each such material is referred to as a "bucket"; the

complete set of materials is shown in Table 1. In order to study them using the UN Comtrade data, we identify Harmonized System (HS) codes for all these materials, at the HS6 level.

We briefly describe the approach in [33] here. At its core, the system uses a directed hypergraph structure to accurately model real-world chemical and manufacturing processes. This format allows for multi-input, multi-output relationships, capturing the complexity of industrial transformations while maintaining full traceability from raw materials to end products. The network grows through an iterative bottom-up discovery process, beginning with user-specified seed materials (such as boron, cobalt) and expanding outward to identify new intermediates, processes, and downstream technologies. Importantly, every pathway is anchored back to naturally occurring materials, ensuring upstream traceability.

To validate, the system employs large language model (LLM) agents that classify materials, identify transformation processes, and score the quality of references. These agents are supported by tools like the Model Context Protocol (MCP), which gives them structured access to HS code databases, online resources, and a continuously updated reference store.

Validation is central to the pipeline: only processes backed by high-quality, relevant sources are retained. Less credible or poorly supported entries are automatically removed to preserve the integrity of the network. Over time, the system builds a semantic cache, allowing it to become smarter and more context-aware with each iteration.

The final output is a directed hypergraph where each node represents a material (classified using Harmonized System codes), and each edge represents a transformation process, supported by reference data. As the network evolves, it provides a transparent and traceable framework to understand technological dependencies. In this analysis we only use the primary forms of the target materials produced using this approach.

4.2.2 Network models and metrics

The Comtrade dataset can be represented as a directed temporal network. In our multiplex network analysis, a multiplex network [10] over a set B is defined as $\mathcal{G} = \{G_i : i \in B\}$, where $G_i = (V, E_i, f_i)$ is the network in the ith layer, V denotes the set of countries, E_i corresponds to the trade flows associated with commodity i, with $(x, y) \in E_i$ if there is an export from x to y with value $f(x, y, i, v)$, and quantity $f(x, y, i, q)$.

Each layer corresponds to a commodity, which is defined by a 6-digit HS code. This network is built using the UN Comtrade database; the raw data is filtered to include relevant commodities, and transactions are aggregated by years. Within each commodity layer, a directed and weighted edge from one country to another signifies that the exporting country supplies that commodity to the importing country, with the edge weight stating the traded value or traded quantity. We construct multiplex networks for each bucket in Table 1.

4.2.3 Metrics for disruption analysis

We study three kinds of metrics associated with trade disruptions.

Disruption score and maximum disruption sets. We define the disruption score of a set $S \subset V$ of countries for a commodity i, denoted $g_i(S)$, as the loss in trade flow (which can be defined in terms of value or quantity) when the set S is blocked. Formally, for flow values,

$g_i(S) = (\sum_{(x,y)\in E_i,\{x,y\}\cap S\neq\emptyset} f(x,y,i,v)/(\sum_{(x,y)\in E_i} f(x,y,i,v))$. We also consider the disruption scores restricted to imports to the US and China. We use $g_i(x,S)$ to denote the score for country x, which is defined as $g_i(x,S) = (\sum_{(y,x)\in E_i, y\in S} f(y,x,i,v)/(\sum_{(y,x)\in E_i} f(y,x,i,v))$; flow disruptions for quantities are similarly defined. We extend this to the multiplex network for a set B of commodities, and define $g(S) = \max_{i\in B} g_i(S)$ as the maximum reduction in flow, across all commodities in B. When $S = \{x\}$ consists of a single country x, we refer to $g_i(\{x\})$ as the *disruption score* for country x. For sets of size k, we define the *maximum disruption set*, $S^*_{k,i} = \arg\max_{|S|=k} g_i(S)$ and maximum k-disruption score $D(i,k) = g_i(S^*_{k,i})$; we consider $k = 1,2,3$, and drop the commodity subscript when it is clear from the context.

The solutions S^*_k need not be consistently increasing or decreasing (monotone) with respect to k, i.e., it is possible that $S^*_k \not\subset S^*_{k'}$ for $k < k'$. Importantly, a country included in S^*_k is not necessarily included in $S^*_{k'}$ for $k' > k$, nor does its influence remain constant across these sets. To illustrate, consider a country U importing a bucket of commodities B, composed of two layers: A_1 and A_2. In layer A_1, imports are highly concentrated, with country C_1 supplying 80% of total imports and the next largest supplier, C_2, contributing only 5%. In layer A_2, however, C_1 and C_2 respectively account for 50% and 40% of imports. In this scenario, the set S^*_1 would consist solely of C_1, whose removal from layer A_1 results in an 80% disruption. Conversely, S^*_2 might reflect the combined removal of C_1 and C_2 in layer A_2, yielding a 90% disruption, of which only 50% is attributable to C_1.

Leverage ratio. This is motivated by the observation in [12] that US sources a large fraction of critical medicines from certain countries (such as Mexico, China), but these account for a small fraction of exports from these countries to the US by value. We define the leverage score of a country y over country x for commodity i as a ratio of the percentage of import of i by x from y (over all suppliers, by quantity) and the percentage of import of i by x from y (over all commodities, by value). Formally, $\ell(x,y,i) = \frac{f(x,y,i,q)/\sum_{y'} f(x,y',i,q)}{f(x,y,i,v)/\sum_{i'} f(x,y,i',v)}$.

Changes in reachability and shortest path distances. We define $d(x,y,i,S)$ as the shortest path distance from country x to country y, in the trade network restricted to commodity i, when set S of countries are disrupted. Let $n(x,i,S)$ denote the number of nodes which can reach country x for commodity i when S is disrupted. We define $d_{avg}(x,i,S) = \sum_y d(y,x,i,S)/n(x,i,S)$ denote the average distance for country x, restricted to reachable nodes; this is undefined when no suppliers are connected to x.

5 Related work

As mentioned earlier, use of network models and analyses for global trade are active areas of research. We summarize the datasets which have been used, and some of important types of analyses which have been done.

Datasets. One of the most commonly used dataset is from UN Comtrade [4], which provides very detailed data on transactions between countries; this includes information on the commodity using the Harmonized System (HS) codes, dates, quantity and value. Given this is based on reported data, it can have inconsistencies and discrepancies between importer and exporter reports. The BACI [19] dataset is constructed from the UN Comtrade, and tries to reconcile the discrepancies between export and import reports.

Economic information of different types is commonly used in trade network analysis. A number of national and international agencies, such as the National Bureau of Economic Research (NBER) [3], and WTO Stats [5] and the International Monetary Fund (IMF) collect and distribute such data. Another type of data that is used extensively is related to supply chains, in the form of multi-region input-output table (MRIO) models, such as Global Trade Analysis Project (GTAP) [2] and Eora [1]. Such datasets split the economy into sectors, and provide estimates of trades between sectors across countries.

Network analyses. A lot of work in this area, e.g., [13, 20, 23, 26, 27, 30, 31], has been been focused on country level trade networks, e.g., those constructed from Comtrade or BACI datasets. Liu et al. [27] provide a survey of the kinds of analyses done on trade networks.

Different kinds of topological properties have been considered in prior work. Trade networks are directed, weighted and temporal. An important type of analysis is based on centrality, which is commonly used in network science to formalize the topological role, position and influence of nodes. Many types of centrality have been introduced based on the dataset, and the type of dependencies considered; these include degree centrality, closeness centrality (defined based on distances), betweenness centrality (quantifying the fraction of shortest paths passing through a node), and eigenvector centrality (using the largest eigenvector), e.g., [25, 26, 35]. Clustering coefficient (which quantifies the fraction of neighbors of a node which are connected) has also been studied in trade networks, but has to be extended for directed networks [15]. Distributions of distances between countries in the trade networks have been used for understanding and prediction of trade relationships [16, 34]. Correlations between structural properties have been used in further understanding trade relationships and robustness of the network. For instance, correlations between centrality and clustering have been used to formalize the fragility of trade networks [26, 31]. Korniyenko et al. [26] combine these metrics with a notion of substitutability (which differs from the one we study) to evaluate product fragility on global trade networks for different commodities. The focus of this work has been on the characteristics of the trade network as a whole rather than on the risks for specific countries.

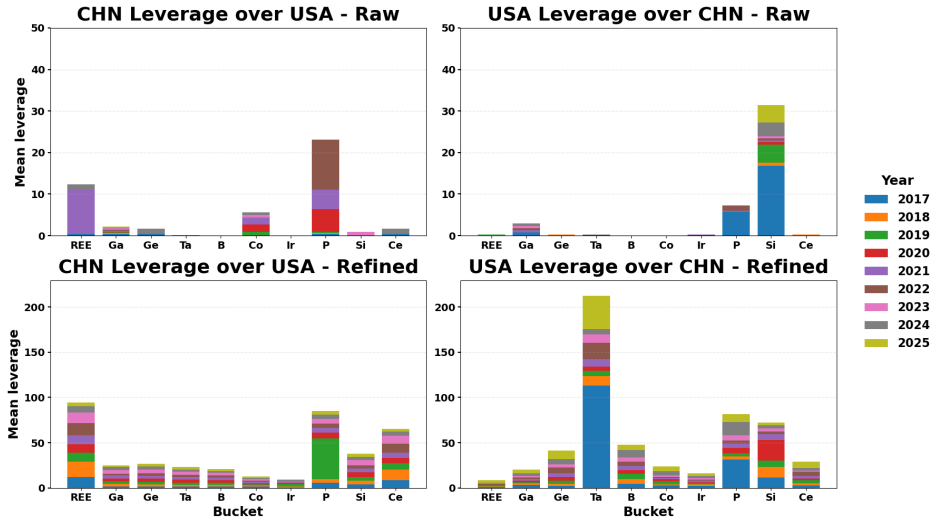

Figure 9: US and China leverage ratios over time.

6 Appendix

6.1 Temporal leverage ratios

Figure 9 shows the relative leverage ratios for US and China for each bucket, and over time. It shows that among raw materials China exhibits leverage in REE, Cobalt and Phosphorus. REE leverage is persistent with spikes in specific years. For most other raw materials China's leverage is low or near zero. US leverage over China in raw materials is generally weak, except for silicon and phosphorus, where leverage increased sharply in some years.

The temporal variation in leverage ratios highlights that strategic dependence is not static, but evolves with market conditions. Materials exhibiting persistent leverage across multiple years, such as China's leverage over the U.S. in refined REE and phosphorus, or U.S. leverage over China in refined tantalum, reflect structural vulnerabilities that cannot be addressed through short-term adjustments or market rebalancing alone. These cases warrant long-horizon policy responses, and sustained investment in domestic processing capacity. Continuous monitoring of leverage over time allows policymakers to distinguish between structural dependencies that require systemic interventions, and temporary bottlenecks that can be mitigated through measures like stockpiling, demand management etc.

6.2 Leverage ratios of coalitions

In the absence of coalitions, China holds a significant leverage advantage over the US in several raw materials, including cerium (Ce), phosphorus (P), cobalt (Co), germanium (Ge), and rare earth elements (REE). The leverage ratios for these raw materials improve for the US when coalitions are formed, as shown in Figure 10. For instance, a coalition between the US and Australia is sufficient to eliminate China's leverage in cerium, phosphorus, and germanium; however, China retains a comparative advantage in cobalt and REE.

Similarly, a bilateral coalition between the US and Germany eliminates China's leverage in phosphorus, while cooperation with Canada substantially reduces China's leverage in raw REE. Nonetheless,

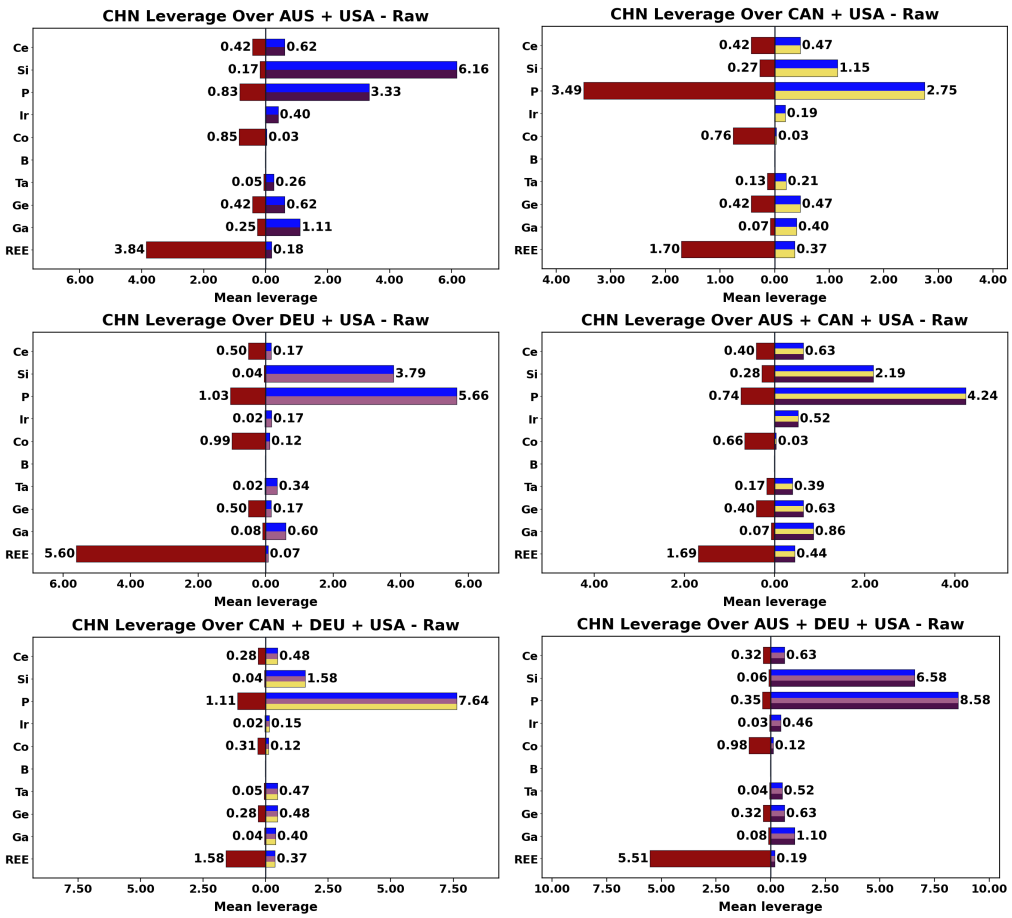

Figure 10: Leverage ratios of China over the coalitions are in red (to the left of the origin) and for the coalitions of US with subsets of Canada, Australia and Germany, over China, are in multicolors (to the right of the origin) for different buckets (on the y-axis) of raw materials. LRs are calculated for each year and for each bucket, and then reported as the bucket-level mean across years (2017-2025). The coalitions are: Top left: US, Australia; Top right: US, Canada; Middle left: US, Germany; Middle right: US, Canada, Australia; Below left: US, Canada, Germany; Below right: US, Australia, Germany.

even a four-country coalition comprising the US, Australia, Germany, and Canada is insufficient to fully offset China's leverage in raw REE and raw cobalt.

In the case of refined materials, the leverage ratios achieved by coalitions vis-a-vis China are substantially higher than those obtained by the US acting alone. For instance, a bilateral coalition between the US and Germany is sufficient to eliminate China's leverage across all materials except rare REEs. A similar outcome is observed when the US forms a coalition with Australia. Moreover, a trilateral coalition comprising the US, Australia, and Germany fully neutralizes China's leverage, including in refined REEs.

6.2.1 Temporal leverage ratios of China versus the coalition

In figure 12 we analyze the leverage ratios of the coalition (US, Canada, Australia, Germany) versus China, over time to see whether dependencies on commodities are permanent or temporary, and

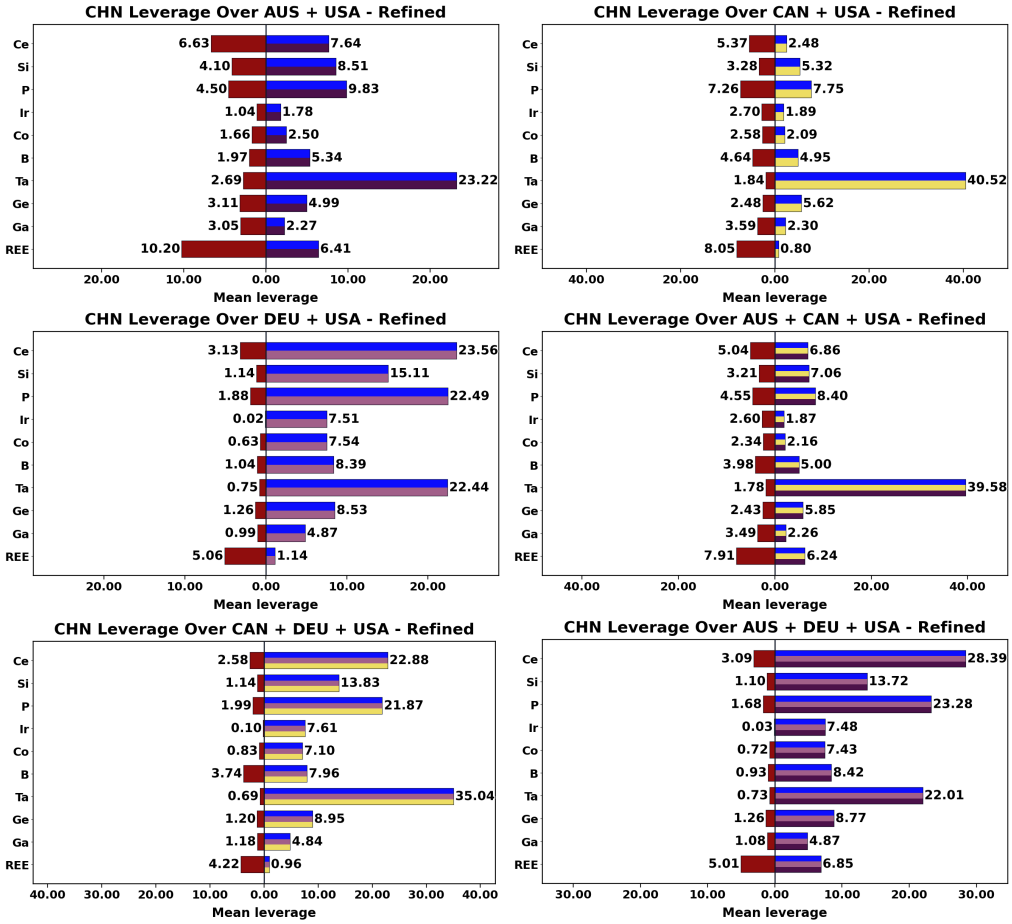

Figure 11: Leverage ratios of China over the coalitions are in red (to the left of the origin) and for the coalitions of US with subsets of Canada, Australia and Germany, over China, are in multicolors (to the right of the origin) for different buckets (on the y-axis) of refined materials. LRs are calculated for each year and for each bucket, and then reported as the bucket-level mean across years (2017-2025). The coalitions are: Top left: US, Australia; Top right: US, Canada; Middle left: US, Germany; Middle right: US, Canada, Australia; Below left: US, Canada, Germany; Below right: US, Australia, Germany

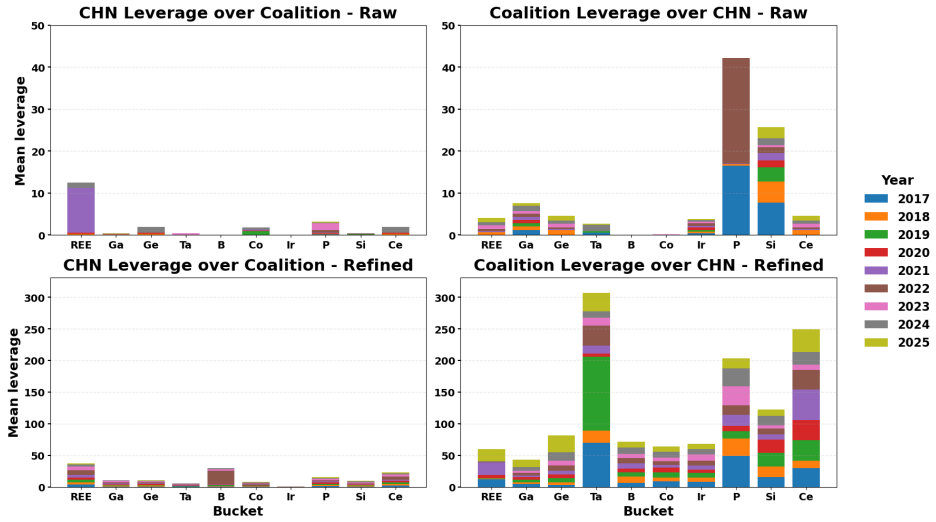

Figure 12: Temporal leverage of a coalition (US, Australia, UK and Canada) and China. Anil, this should be Germany not UK, right?

if new vulnerabilities are emerging. Results show that across years, raw-material leverage remains relatively persistent for China, especially in REE and phosphorus, whereas refined-material leverage shifts much more strongly toward the coalition.

Some buckets show persistent leverage across nearly all years. For example, refined tantulum, phosphorus and celium exhibit high coalition leverage every year after 2018, which is indicative of durable supply-chain control. However, even when coalitions form, China retains raw-stage leverage in REE, cobalt and phosphorus which is indicative of geological concentration and barriers to entry.

Overall, temporal patterns reveal that coalition strategies are most effective and increasingly so over time in case of refined materials, but persistent leverage of China in raw-materials reflects structural constraints that cannot be resolved through alliances alone.